

CSCI-UA 102 — Quiz 3

Recursion

Name: _____ NetID: _____

funNum

For an integer array of size N , I define the **funNum** of the array as follows:

- If the array has **no elements**, then funNum is **0**.
- If the array has **one element**, then funNum is equal to **that element**.
- Otherwise, funNum is equal to the **sum** of:
 1. the last element of the array,
 2. the funNum of the first $N - 1$ elements, and
 3. the funNum of the first $N - 2$ elements.

Examples:

- $\text{funNum}(\{\}) = 0$
- $\text{funNum}(\{5\}) = 5$
- $\text{funNum}(\{5, 3\}) = 3 + 5 + 0 = 8$
- $\text{funNum}(\{5, 3, 1\}) = 1 + 8 + 5 = 14$

Write an efficient, recursive method:

```
public int funNum(int[] array)
```

What is the time complexity of your solution? _____